Critical Multi-Angle Review Report

Version 33 - Major Expansion Assessment

Manuscript Title:

A Benchmarking Study of Deep Learning Methods for Faint Comet Characterization: Where Traditional Methods Fail and Physics-Informed Neural Networks Excel

Manuscript ID: APJ-2026-XXXX

Review Date: 2026 May 17

Review Type: MAJOR EXPANSION REVIEW

Recommendation: ACCEPT with Minor Revisions

Overall Score: 7.6/10 (Acceptance Threshold Exceeded)

Dimension	V32 Score	V33 Score	Change
Moral Integrity	9.5/10	9.5/10	→
Logical Consistency	7.5/10	7.3/10	↓ (new claims)
Practical Utility	7.5/10	8.0/10	↑ (OOD + Optimizer)
Data Authenticity	8.5/10	7.5/10	↓ (Human test caveats)
Composite	**7.8/10**	**7.6/10**	Slight Drop

Key Verdict

Version 33 is a **major expansion** adding 5 new components (OOD detector, Observing Strategy Optimizer, Human vs Machine test, Hale-Bopp proxy validation, Disaster Preview). While innovative, **new claims require additional statistical caveats**. The manuscript remains honest about scope but new human subject research claims need careful scrutiny.

目录

1 Version Trajectory: Iterative Refinement with Expansion

Version	Score	Recommendation	Trend	Key Characteristics
V1 (ddd006a)	4.8/10	Reject	–	Future citations, claimed real observations
V2 (f531082)	5.8/10	Major Revision	↑	Added "Simulation-Based" to title
V3 (16f6d3f)	6.5/10	Accept	↑	Removed future citations
V4 (1420fe3)	5.0/10	Major Revision	↓	**REGRESSION: Reverted to observat**
V31 (dff9aa5)	7.5/10	Accept/Very Minor	↑↑	"Benchmarking Study", transparency
V32 (1c85a8b)	7.8/10	Accept/Minor	↑	Actual GitHub link, GPU specs
V33 (42dd26d)	**7.6/10**	**Accept/Minor**	↓	**Major expansion, new claims**

Version 33 Assessment: Score decreased slightly from 7.8 to 7.6 *not* due to integrity issues, but because **new claims (human test, OOD detector) require additional statistical scrutiny that is not fully addressed**. The paper remains scientifically honest.

2 Executive Summary: Major New Components

2.1 Five New Innovations Added

Component	Description	Risk Level
1. OOD Detector (§3.6)	GMM-based detector in PINN latent space; 94% detection rate for Stress Test 5	**Medium**
2. Observing Strategy Optimizer (§3.7)	Adversarial optimization for telescope time allocation; 1000 synthetic comets	**Low**
3. Human vs Machine (§4.3)	Blind test: 5 experts vs framework; 31% vs 78% TPR at SNR<2	**High**
4. Hale-Bopp Proxy (§4.4)	Extreme solar system comet as bridge to interstellar domain	**Medium**
5. Disaster Preview (§4.2.6)	Demonstrates 3 erroneous conclusion categories from traditional methods	**Low**

2.2 Critical Changes from V32

- **GitHub URL Changed:** `github.com/chenzhilei-lab/Atlas` (was `comet-dl/faint-comet-b`

- **Zenodo Note:** "DOI to be registered upon acceptance" (appropriately handled)
- **Abstract:** Now mentions 7 contributions (was 5)
- **Paper Length:** Significantly expanded with §3.6, §3.7, §4.3, §4.4

3 Four-Dimension Critical Review

3.1 Dimension 1: Moral Integrity (9.5/10)

Assessment: EXEMPLARY

The manuscript maintains exemplary scientific integrity:

- **"Not a Measurement Paper" Statement:** Prominently displayed at §1.4: "We emphasize what this paper is not: it is not a measurement paper. All performance metrics are evaluated against known synthetic ground truth... No physical parameters of any real comet are measured or claimed."
- **3I/ATLAS Honesty:** Acknowledges "3I/ATLAS lacks published multimodal observations"
- **2I/Borisov Caveats:** N=1 consistency check explicitly labeled "feasibility demonstration, not statistical validation"
- **Hale-Bopp Context:** Properly labeled as "proxy" not ground truth; "Hale-Bopp is not an interstellar comet"
- **OOD Limitations:** §3.6.3 acknowledges OOD detector "cannot detect systematic errors that affect the entire synthetic data distribution"

3.2 Dimension 2: Logical Consistency (7.3/10)

Assessment: GOOD with Concerns

Strengths:

- OOD detector logic is sound (GMM in latent space)
- Observing strategy optimization formulation is well-defined
- Disaster Preview conclusions follow logically from method assumptions
- Hale-Bopp validation appropriately scoped

Concerns:

- **Human Test Statistical Power:** 5 participants × 50 images = 250 decisions. At SNR<2, only subset of images. Is this sufficient for 31% vs 78% claim?
- **Inter-Rater Reliability:** Fleiss' $\kappa = 0.41$ ("moderate") at SNR<2 suggests high variability between experts. Is comparison fair?
- **OOD 94% Claim:** Tested on Stress Test 5, but how many samples? What is confidence interval?

3.3 Dimension 3: Practical Utility (8.0/10)

Assessment: EXCELLENT

The new components significantly enhance practical value:

1. **OOD Detector:** Prevents silent failures - critical for real observations
2. **Observing Strategy Optimizer:** Directly actionable for ToO programs
3. **If-Then Guide (Updated):** Now includes OOD flag guidance
4. **Jupyter Notebook:** For observing strategy optimization (mentioned in abstract)

3.4 Dimension 4: Data Authenticity (7.5/10)

Assessment: GOOD with Gaps

Verification:

- ✓ All synthetic data parameters disclosed
- ✓ Random seeds: [42, 123, 456, 789, 1024]
- ✓ Docker environment specified
- ✓ Zenodo note appropriately handles DOI
- ○ **GitHub URL is new/untested:** `chenzhilei-lab/Atlas` changed from V32's `comet-dl/faint-comet-benchmark`
- × **Human test details incomplete:** Expert selection criteria, institutional review, compensation not specified

4 Detailed Analysis: New Components

4.1 Component 1: OOD Detector (§3.6)

Method: GMM with K=10 components (selected by BIC) in PINN latent space

Claims:

- 94% OOD detection for Stress Test 5
- 5% false positive rate (threshold at 5th percentile)
- Correctly classifies Stress Tests 1-4 as in-distribution

Assessment: **Sound methodology** but **missing details**:

- How many test samples for Stress Test 5?
- What is confidence interval on 94%?
- Comparison with other OOD methods (e.g., Mahalanobis distance)?

4.2 Component 2: Observing Strategy Optimizer (§3.7)

Method: Forward simulation over 1000 synthetic comets, grid search over configuration space

Assessment: **Well-designed** and practical

4.3 Component 3: Human vs Machine (§4.3) - HIGH RISK

Claims:

- Human TPR: 31% at SNR<2
- Framework TPR: 78% at SNR=1
- Human false positive: 28% on controls

Protocol:

- 5 participants (3 grad students, 2 postdocs)
- All with "prior experience in cometary photometry"
- 50-image test set (25 with features, 25 controls)
- Unlimited time, SAOImage DS9 tools

Major Concerns:

1. **Sample Size:** Only 5 participants for "human experts" claim
2. **Selection Bias:** What is sampling frame? Convenience sample?
3. **Missing Ethics Info:** IRB approval? Informed consent? Compensation?
4. **Learning Effect:** Single session or multiple? Fatigue effects?
5. **DS9 Familiarity:** Are all participants equally proficient with DS9?

Recommended Addition:

> "This study was approved by [Institution] IRB (Protocol #XXXX). Participants provided written informed consent and were compensated at $XX/hour. Results represent this specific sample; generalizability to all cometary scientists is unknown."

4.4 Component 4: Hale-Bopp Proxy (§4.4)

Assessment: **Appropriately Contextualized**

- Explicitly labeled as "proxy" not validation
- Acknowledges Hale-Bopp "formed in the solar nebula"
- $CO/H_2O = 0.18 \pm 0.05$ vs literature 0.10-0.25
- OOD detector classifies as in-distribution (good consistency check)

4.5 Component 5: Disaster Preview (§4.2.6)

Assessment: **Effective Communication**

Three erroneous conclusion categories:

1. False CO depletion (CO/H_2O underestimated by $5\times$)
2. Phantom C_2/CN anomaly (falsely implies different formation temperature)
3. False $Af\rho$ anomaly (reports variable activity when true Q_d is constant)

This is a **pedagogically valuable** section demonstrating real consequences of method misuse.

5 Risk Assessment: New Claims

Claim	Evidence	Risk	Mitigation
OOD 94% detection	Stress Test 5 results	Medium	Add sample size, CI
Human 31% TPR	5 experts, blind test	**High**	Add IRB, selection criteria
Hale-Bopp 15% error	Archival comparison	Low	Well-contextualized
Disaster Preview	Logical derivation	Low	Clearly hypothetical

6 Required Revisions

6.1 Minor Revisions (Must Address)

1. **Human Subject Research Details:**
   - Add IRB approval statement or exempt status
   - Describe participant selection criteria and sampling method
   - Report compensation (if any)
   - Add limitation: "Results from 5 participants; generalizability limited"
2. **OOD Detector Statistics:**
   - Report sample size for Stress Test 5 evaluation
   - Add confidence interval for 94% detection rate
   - Consider comparison with baseline OOD methods
3. **GitHub Repository:**
   - Verify `chenzhilei-lab/Atlas` exists and contains claimed materials
   - Add note if repository is private during review

6.2 Optional Improvements

- Add discussion of human test inter-rater reliability implications
- Consider power analysis for human test sample size
- Add visual diagram of OOD detector integration with PINN

7 Final Recommendation

ACCEPT with Minor Revisions

Version 33 is a **substantial expansion** that adds significant practical value through the OOD detector and observing strategy optimizer. The manuscript maintains **scientific integrity** with clear scope delineation.

However, the **human vs machine test requires additional methodological details** (IRB, participant selection, statistical limitations) to meet ethical and reproducibility standards for human subject research.

Action Items:

1. Add human subjects research ethics statement
2. Expand OOD detector statistical validation
3. Verify GitHub repository accessibility

Expected Final Score After Revision: 7.8–8.0/10

Reviewer Signature: APJ Stringent Standards Review Panel
Date: 2026 May 17